

Questions for Materialists

Post date [2026-08-05](https://shaltiel.org/wp/questions-for-materialists/). Watch at <https://shaltiel.org/wp/questions-for-materialists/>

What is the signature of wisdom? A purposeful arrangement of parts—for where parts are ordered to an end, intelligence is implicated.¹

See: [Cosmic Accident or Intelligent Design](#) and [The Improbable Mathematical Odds Against Evolution](#) (D. Axe, 18m).



If you don't believe that **intelligent design** was involved in the origin of life, then you essentially believe that a few unintelligent forces of physics acting alone could, under the right conditions, cause life to arise from dead chemicals on barren planets and eventually produce human intelligence and civilization. Peter Urone, for example, claims in his 2001 textbook *College Physics* that "only four distinct forces account for all known phenomena."

As it turns out, those four forces are very cleverly designed and **fine-tuned** for life. That itself requires an explanation.

The contrast is striking. The Moon and other planets are barren wastelands, hostile to life. By contrast, Earth is a habitat remarkably suited to carbon-based life. It possesses an extraordinary, interconnected system of oceans, atmosphere, sunlight, clouds, winds, rainfall, rivers, fertile land, and continuously recycled fresh water. The [water cycle](#) is only one component of a much larger system that provides a stable habitat for carbon-based intelligent life and, ultimately, human civilization. Water itself has a remarkable combination of properties suited to biological life, while Earth's atmosphere, hydrosphere, and other physical conditions work together to sustain it. How did this extraordinarily coordinated, life-sustaining system arise from those four forces?

However, although those four forces (such as gravity and electromagnetism) are necessary for life, they are not on their own sufficient to explain the origin of the **coded information** needed for the **nano-machinery and organized complexity of cellular life**. And they are certainly not clever enough, by themselves, to explain how human intelligence and civilization arose on Earth.

Four Questions for Materialists

Here are the major questions materialists must answer to explain how human intelligence and civilization could have arisen on Earth by undirected forces that did not have us in mind.

1. **(Origin of life)** How did first life arise from dead chemicals? How does all that nano-machinery and coded information in the cell arise from molecules that do not care about life?
2. **(Reproduction)** How are these living things able to self-replicate, passing their current complex structures on to their descendants, generation after generation, without significant degradation?
3. **(Evolution)** How did the accumulation of replication errors (random mutations) over millions of years result in the organized complexity of plants and animals?
4. **(Evolution of intelligent humans)** How did this process of evolution eventually produce intelligent, conscious, humans, who are able to design computers and spaceships and smart phones?

[A Mathematician's View of Evolution](#) (Granville Sewell, 7m)

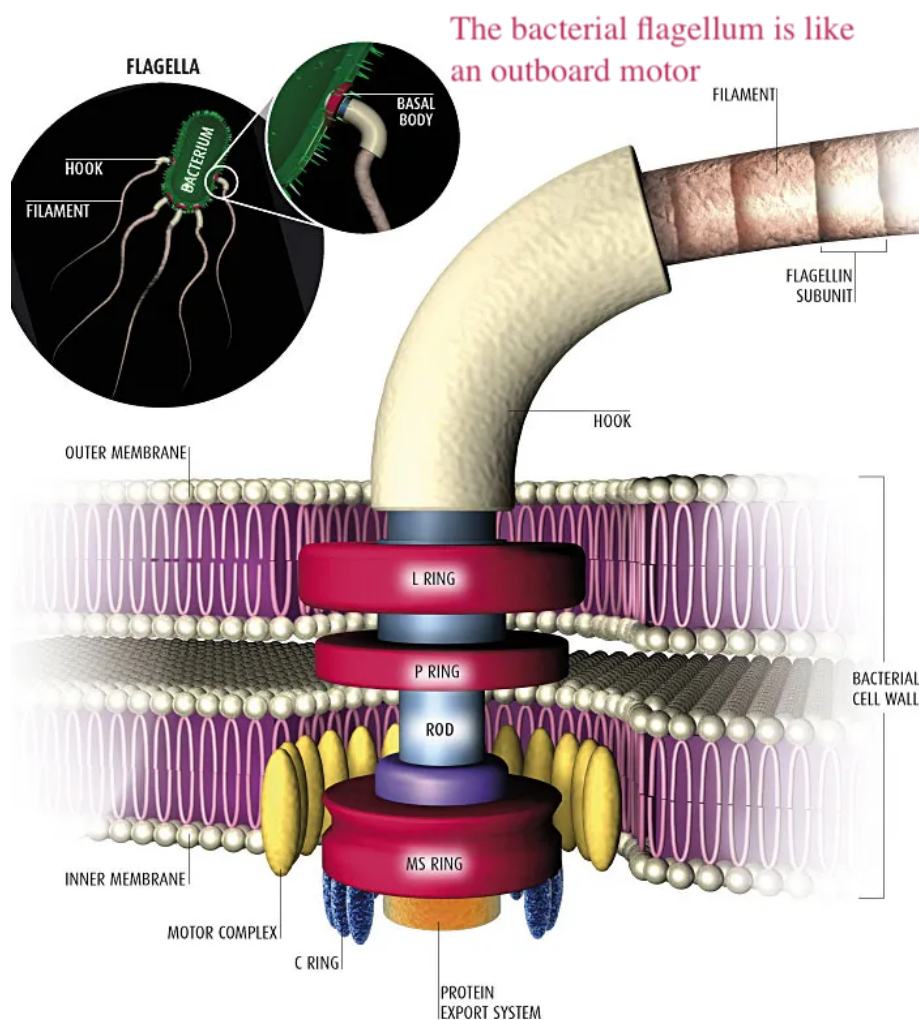
Minds and Purpose

Minds (and *only* minds) can have purposes. Thus, to the extent it can manipulate things, a mind can arrange parts to achieve its purposes. Of course, we ourselves have minds. And it is a fundamental power of mind that it can discern purposes. Thus, we can recognize that a mind has acted by perceiving a **purposeful arrangement of parts**. How else would we recognize another mind.

For the purpose of detecting other minds, "parts" can be virtually anything. Examples include: the purposeful arrangement of sounds in speech; words and letters in writing; mechanical parts in machinery.

Likewise, the *purpose* of the heart is to pump blood, the eyes to see and wings for flight (read any textbook on anatomy and physiology).

Other phrases that people use to indicate intelligent design all boil down to purposeful arrangements of parts. For example, Stephen Meyer likes to point out that we know intelligent agents produce information, so when we come across **coded information in a computer program**, we can conclude it was produced by an intelligent agent. Likewise for the coded information in the simplest cell. True enough. Yet how do we know there is information in a string of zeroes and ones — in a computer program? Only if we find that they are arranged for a purpose; that is, if the computer program has a function, if it can do something purposeful.



In the same way, Michael Behe (Darwin's black box) describes how **irreducibly complex systems** such as the **bacterial flagellum** resist Darwinian explanation. But how do we know they are designed? Because we see they can do something, that they have a purpose, they are a purposeful arrangement of parts. Irreducibly complex systems have two relevant properties — their discontinuous nature resists Darwinism and their manifest purposiveness strongly point to design.

[The Story Behind Darwin's Black Box](#) (Michael Behe, 12m)

The [eye](#) demonstrates "**specified complexity**." The "specified" in the phrase *specified complexity* is pretty much the same as "purposeful," and "complexity" the same as "arrangement." So, the eye is a "purposeful arrangement" of parts.



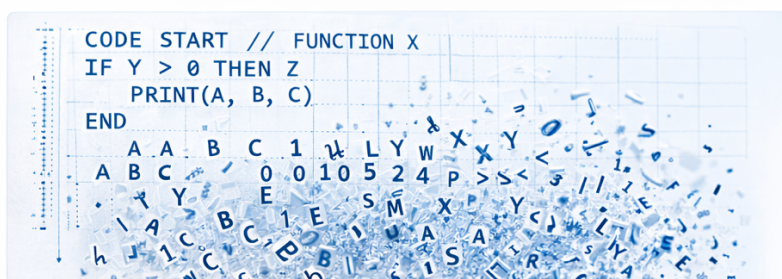
Above, **Plan and Purpose**: An Eastern Bluebird in flight. Its **wings** are a highly integrated aerodynamic system of bones, joints, muscles, and feathers. Its **eyes** allow it to hunt visually from perches and detect tiny prey on the ground from 60 feet or more away, then calculate and

execute the flight needed to intercept it. Its tiny **brain** must continuously integrate vision, balance, wing position, airspeed, obstacle avoidance, and landing control. In migratory populations, its brain supports seasonal orientation and navigation over very large distances. Northern Bluebirds do not simply wander south; their seasonal movements between breeding and wintering grounds require precise orientation, timing, and migration over distances of 2,000 miles.

Origin of Life

Information Enigma

Despite intensive research ever since the **Miller-Urey experiment (1953)**, scientists remain clueless: how did the first **information-rich, self-reproducing cell** arise from dead chemicals?



Random Mutation

When we randomly change alphabetic **text** or computer code, the **universal result** is the **degradation of meaning** and the generation of **gibberish**.
(As noted by MIT's Murray Eden).

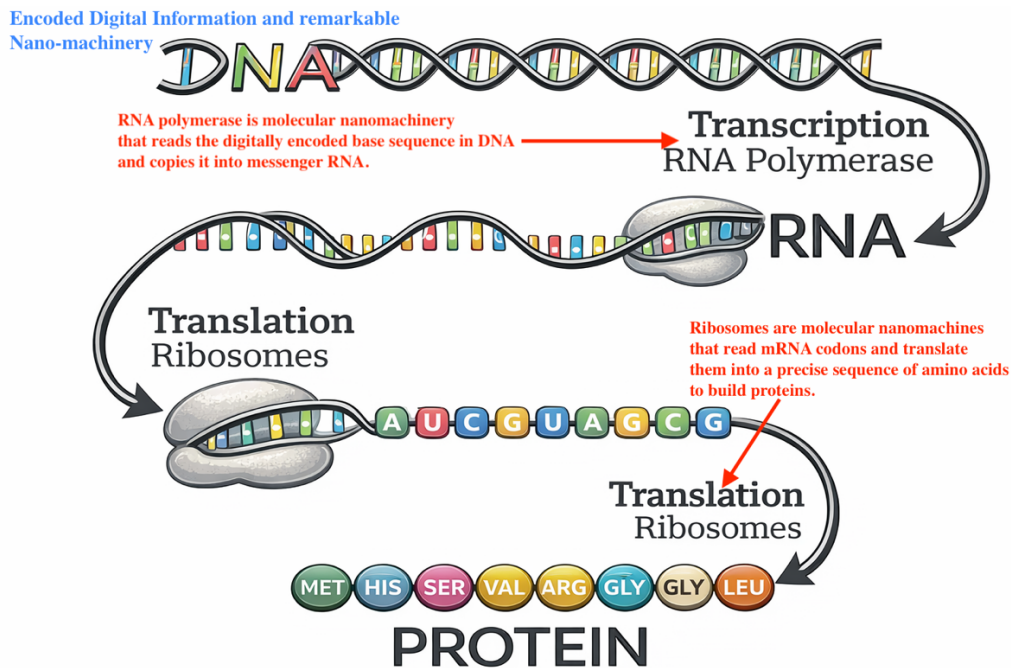


Information, Not Just Chemistry

These **sequences** are **not random chemical reactions**. As discovered by **Watson and Crick (1953)**, the chemical constituents in DNA function exactly like letters in a written language or digital symbols in a section of software code.

At Harvard in 2023, Rice University chemist James Tour challenged researchers in origin-of-life chemistry to explain, at the level of actual chemistry, how unguided prebiotic processes could produce the components and organization required for the first living cell. Tour had invited ten leading origin-of-life researchers to engage his objections.

Lee Cronin, professor of chemistry at the University of Glasgow, was the only one who accepted the invitation to respond publicly. As Tour argued in their exchange, Cronin's assembly theory failed to provide a demonstrated chemical pathway from non-living matter to the first living cell. In Tour's words, when it comes to explaining the origin of life, **we remain "clueless."** [Clueless on origin of the cell](#) (James Tour, 6m)



Inference to the Best Explanation

Premise 1	Despite extensive investigation, no unguided materialistic processes have been discovered with the causal power to produce the purposeful arrangements of parts necessary to construct the first cell and subsequent biology. *
Premise 2	By our common universal experience, intelligent agency alone is known to possess precisely this causal power to arrange parts purposefully to an end.
Conclusion	Intelligent design constitutes the only known causally adequate explanation for the origin of the purposeful arrangement of parts in the cell and subsequent biology.

* Chemists and biologists remain clueless. See footnote for the problem of the combinatorial explosion of the search space.

Reality Check: the finger of God

There is enormous hype from the scientific establishment about the supposed creative power of chemical and biological evolution. Yet the history of molecular biology has made the problem harder, not easier. The first living cell once seemed comparatively simple. As our knowledge has increased, however, the minimum requirements for life have proved to involve astonishingly intricate systems of coded information, molecular machines, membranes, energy production, regulation, error correction, replication, and mutually dependent biochemical pathways. With every new layer of complexity discovered, the explanatory goalposts recede further, and the combinatorial information problem becomes more formidable.

As science becomes more knowledgeable about how life works the problem of explaining how nonliving chemistry could have produced the first organized living system without plan or purpose becomes increasingly difficult. The goal post of understanding is receding!

If you want a reality check, consider the third plague to remove the brutal slavery imposed by Egypt and its Pharaoh on the helpless Israelites (Ex. 8.12-15):

וַיֹּאמֶר ה' אֶל־מֹשֶׁה אָמַר אֶל־אַהֲרֹן נִטֵּה אֶת־מִטְּהָ וְהָדָה אֶת־עָפָר הָאָרֶץ וְהָיָה לְכֹנָם בְּכָל־אֶרֶץ מִצְרָיִם:

Then God said to Moses, "Say to Aaron: Stretch out your rod and strike the *dust* of the earth, and it shall turn to *lice* throughout the land of Egypt." [Living lice from dead chemicals.]

וַיַּעֲשׂוּ־כֵן וַיִּט אַהֲרֹן אֶת־יָדוֹ בְּמִטְהוֹ וַיִּדָּה אֶת־עָפָר הָאָרֶץ וַתְּהִי כֹנָם בְּאֶדָם וּבְבִהֵמָה כָּל־עָפָר הָאָרֶץ הָיָה כֹנָם בְּכָל־אֶרֶץ מִצְרָיִם:

Aaron stretched out his arm with the rod and struck the dust of the earth, and lice came upon human and animal; all the dust of the earth turned to lice throughout the land of Egypt.

וַיַּעֲשׂוּ כֵן הַחֲרָטִמִּים בְּלִטְיָהֶם לְהוֹצִיא אֶת־הַכִּנֹּיִם וְלֹא יָכְלוּ וַתְּהִי הַכִּנָּם בְּאָדָם וּבַבְּהֵמָה:

The wizards did the same with their spells to produce lice, but they could not. The lice remained upon human and animal;

וַיֹּאמְרוּ הַחֲרָטִמִּים אֶל־פַּרְעֹה אֲצַבֵּעַ אֱלֹקִים הוּא וַיִּחַזַּק לִב־פַּרְעֹה וְלֹא־שָׁמַע אֲלֵהֶם כַּאֲשֶׁר דִּבֶּר ה':

And the wizards said to Pharaoh, **"This is the finger of God!"** But Pharaoh's heart hardened and he would not heed them, as God had spoken.

Now go the most advanced biochemical laboratory (2026). Provide the scientists with a single-celled bacterium (an organism vastly less complex than a louse). Once the bacterium is dead, all the chemicals are still in place. Ask the scientists to bring it back to life. The most advanced technology is not yet capable of recreating the organized complexity of life from dead chemicals.

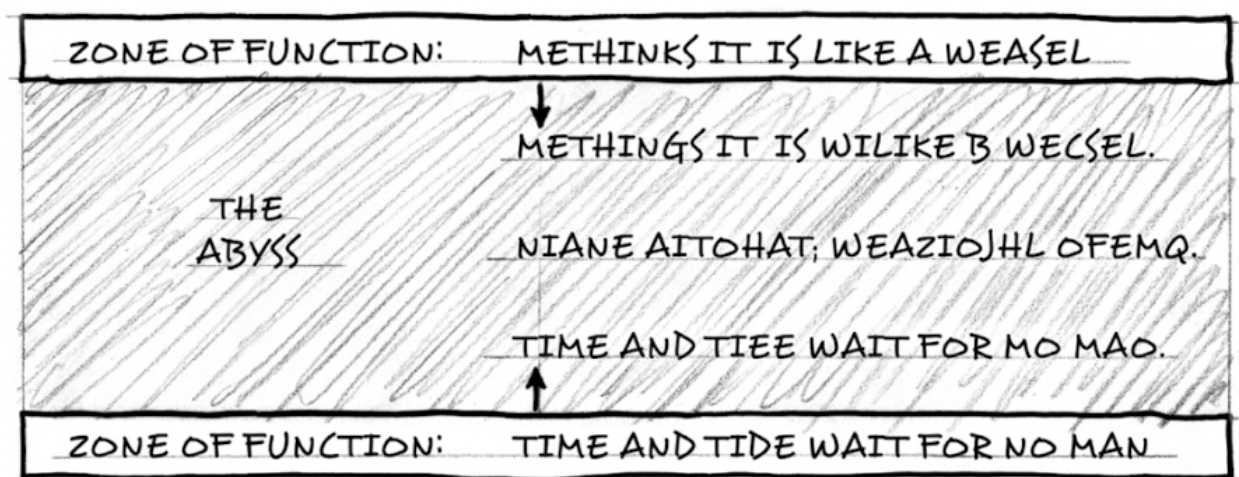
If even intelligent agents armed with modern laboratories cannot yet accomplish this, then what grounds are there for hyped claims about the effortless creative power of unguided chemistry? The goalpost of finding detailed chemical or Darwinian pathways recedes ever further the more we know.

More than 3,300 years ago, the Egyptian science-magicians recognized the *finger of the Creator and Guide* of the universe (the one God, who abhors the injustice of Egypt and its Pharaoh). And so it is today. It is only the stubborn ideology of materialism—the ideology of

the modern Pharaohs—that prevents recognition of a universe buzzing with **Plan and Purpose**.

Footnotes

Fig. 10.3, Stephen Meyer, *Darwin's Doubt*, 2013. See also *Signature in the Cell*, 2009, and [Cosmic Accident or Intelligent Design](#).



1. Many proteins are needed for life. Douglas Axe provided experimental evidence for the rarity of functional protein sequences by estimating how often random amino acid chains can form a working enzyme. For example, the protein *beta-lactamase* contains a sequence of 153 amino acids. For each functional sequence, 10^{77} sequences are not functional. The overwhelming majority of possible sequence combinations are non-functional because they fail to meet the physical and chemical requirements of a working enzyme at three successive levels described in the experiment. So, the functional proteins are small islands in vast spaces of junk.

The rarity of functional configurations makes an appeal to a step-by-step Darwinian explanation of random mutation and natural selection highly improbable. A detailed Darwinian

pathway for the *beta-lactamase* protein must identify a continuous pathway through viable intermediate states—where each intermediate provides a neutral (non-destructive) or functional advantage, despite the fact that the Darwinian mechanism has no *foresight* or goal at its disposal to guide it toward the highly coordinated final system. As the number and interdependence of the required components increase, the space of possible failures expands *combinatorially*, while the burden of specifying such a pathway becomes exponentially greater. Given that there are only an estimated 10^{80} basic particles in the observable universe, the requirement of hundreds of distinct proteins for life presents an extraordinary probabilistic barrier so that the Darwinian blind search mechanism will inevitably fail due to the *combinatorial explosion* of the search space. ↩

Objection: God of the Gaps

What makes this argument different than a G-d of the Gaps argument, or are you in general in favor of arguments which state that G-d is the best answer wherever there is no known natural way to explain a phenomenon?

I am perplexed as to why you think it might be a “God of the Gaps” argument.

The claim is not merely that no unguided explanation has been found despite extensive research—though that alone is reason enough to reject the vast scientific hype, and to reexamine the methodological claims of *scientism*.

But let’s examine the “God of the Gaps” objection carefully.

God of the Gaps Fallacy	
Premise	Cause A cannot produce or explain evidence E.
Conclusion	Therefore, some independent cause B produced or explains E.

The logical error in the “God of the Gaps” fallacy is that the failure of cause A does not by itself provide evidence for some independent cause B. To infer B, one needs either positive evidence that B has the causal power to produce evidence E, or B must be the exhaustive alternative to A. In our argument, both conditions are met. Premise 2 supplies the positive causal evidence.

Inference to the Best Explanation	
Premise 1	Despite extensive investigation, no unguided materialistic processes have been discovered with the causal power to produce the purposeful arrangements of parts necessary to construct the first cell and subsequent biology. *
Premise 2	By our common universal experience, intelligent agency alone is known to possess precisely this causal power to arrange parts purposefully to an end.
Conclusion	Intelligent design constitutes the only known causally adequate explanation for the origin of the purposeful arrangement of parts in the cell and subsequent biology.

This is not an appeal merely to our present ignorance. We have principled reasons for understanding why unguided explanations fail—for example, the combinatorial inflation of the search space and conservation-of-information constraints. We argue from what current operational science demonstrates, not from gaps in the science.

It is like thermodynamics ruling out perpetual-motion machines of the second kind, and the experimental elimination of spontaneous generation. These are not arguments from ignorance, but conclusions based on what we positively know about the relevant systems—through

mathematical constraints and empirical evidence. See the Axe and Dembski references in the footnote below.

That is why the inference is an *abduction*—an inference to the best explanation, one of the most widely used forms of reasoning in science. We seek the *vera causa*—true causes known to operate empirically here and now. Our common universal experience is that purposeful arrangements of parts, wherever their origin can be independently checked, are produced by intelligence. That is positive **causal** evidence stated explicitly in Premise 2.

Premise 2 is positive evidence for the causal powers of intelligent design (= guided processes) via our uniform and repeated experience concerning the cause of specified information, especially communicating digital codes and the irreducible complexity of nano-machinery in even the simplest cell. Thus, the evidence for intelligent design is based on established cause-effect relationships.

It is a **category error** to classify this inference to the best explanation as a “God of the Gaps” fallacy; doing so reflects a fundamental lack of logical clarity about the form of the argument. Furthermore, once the alternatives are stated explicitly as **guided** (exercising intelligence and foresight) **or unguided**, the core inference can also be expressed deductively:

Deduction
Evidence E was produced either by guided or unguided processes.
Evidence E was not produced by unguided processes.
<i>Therefore</i> , E was produced by guided processes.

The footnote (*) explains why Premise 1 in the *abduction* can be strengthened from “despite extensive investigation, no unguided process has been discovered with the required causal power” to the claim that “E was not produced by unguided processes.” On that formulation, the inference can be expressed deductively. I have chosen, however, to retain the abductive version in the main text.

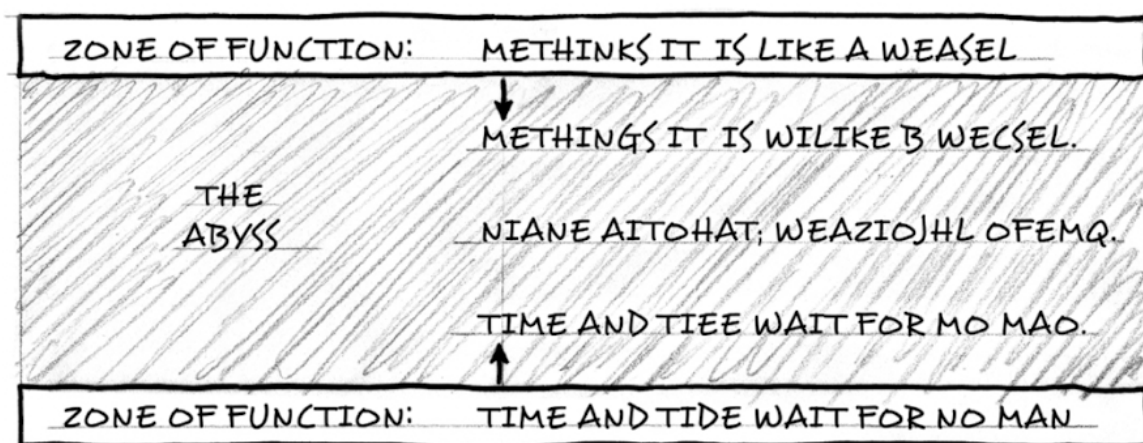
I note that you did not consult the premises of the argument carefully. Had you done so, you would have seen that this is not an argument from ignorance. An objection to an argument must engage its actual premises or the inference from them, rather than substitute a different argument and object to that. Furthermore, “God” does not actually appear in the formal inference; that is a later step. The stated conclusion is that intelligent design is the **causally** adequate explanation. Scientific inquiry rests on common sense realism: that there is an objective world external to the minds of scientists; that this world is governed by regularities of **cause** and **effect** that can be captured in scientific laws; and that the human intellect and perceptual apparatus can uncover and accurately describe these regularities.

Footnote to God of the Gaps Fallacy:

Douglas Axe, “*Estimating the Prevalence of Protein Sequences Adopting Functional Enzyme Folds*,” *Journal of Molecular Biology* (2004);

William A. Dembski, “*The Law of Conservation of Information: Search Processes Only Redistribute Existing Information*,” BIO-Complexity (2025).

Douglas Axe's mutagenesis experiments directly measured how rare functional sequences are within the space of possible amino acid combinations, finding roughly one working sequence among 10^{77} non-functional ones. This is not merely a lack of imagination about possible pathways; it is an experimentally grounded estimate of how sparse the functional regions are within the immense surrounding “abyss” of non-function.



Even granting, for argument's sake, that some detailed Darwinian pathway of any kind—natural selection, co-option, gene duplication, etc.—must somehow be **assumed** to exist, connecting a starting point to the functional target, that assumption would function as a probability **amplifier**, raising the baseline probability p to some workable probability q . But locating such an amplifier is itself a **search for a search** (S4S), governed by the mathematical relation $r \leq p/q$: the probability of finding a search that achieves this amplification is itself tightly constrained by the ratio p/q . The improbability has not been removed, only relocated to a higher-level search. Positing a further mechanism to explain the amplifier only generates a further search for a search (S4S4S), and so on—a regress that never resolves the improbability but merely relocates it (Dembski, 2025, §§11–12).

Dembski uses several intuitive examples to illustrate **Search for a Search (S4S)** and the **displacement fallacy**: an apparently successful search may succeed only because information has been shifted into an unexplained higher-level mechanism rather than genuinely generated by the search itself.

In the **monkey-and-technician** example, a monkey typing randomly can be made to produce Shakespeare if a technician stands behind it and erases every incorrect keystroke. But the difficulty has not been solved; it has merely been displaced into the technician, who must already know the target text (Shakespeare) well enough to distinguish correct from incorrect letters.

Likewise, in Dawkins's **WEASEL** simulation, random mutation is made highly effective by a selective gradient that rewards increasing *similarity* to a pre-specified target phrase “METHINKS IT IS LIKE A WEASEL”, a target unknown to the blind mechanism that has no foresight. The difficulty has merely been displaced not solved. You have to know the target phrase to distinguish correct from incorrect letters. So, you have to know the target to get to it.

The same point is made most explicitly in Dembski's treasure-map example. A blind search of a vast island has a tiny probability p of finding a buried treasure, while a correct map can raise the probability of success to some much larger value q , perhaps even $q = 1$. But then the question becomes how the correct map was found among all the possible maps. That higher-level problem is the search for a search. The map does not eliminate the original improbability; it transfers the informational burden to the task of locating the right map. Dembski's broader claim is that adding still further mechanisms only repeats the process at successively higher levels: the improbability is displaced, not removed.

What is the signature of wisdom? A **purposeful arrangement of parts**—for where parts are ordered and coordinated towards an end, intelligence is implicated.

We see this throughout nature: the water cycle; the *finely tuned* forces and constants of the fundamental laws of nature; the Sun providing the warmth and light required for life; photosynthesis, nutrition, and reproduction; the astonishing molecular machinery of the cell; and integrated organs such as the eye, the mammalian brain, and the wings of birds. Taken together, these coordinated arrangements directed towards life point beyond themselves to an awesome intelligence. No human power can replicate them, let alone originate them. Nor can their explanatory force be dismissed by forcing them into a purely *materialistic* straitjacket.

We should be **singing in the rain** and listening with wonder when the birds begin to chirp as the Sun emerges—for these ordinary things are themselves extraordinary manifestations of an intelligible and purposeful order of goodness. See *Gate of Reflection*, in *Duties of the Mind*.

“Darwin introduced *materialism* into nature, with his theory of evolution. Marx introduced *materialism* into history with his theory of “historical materialism.” Freud, with his insistence that man's actions are entirely predetermined by his previous experiences, introduced *materialism* into the very soul of man. All these types of *materialism* amount to a *churban* of *kedushah*. Freud's materialism was particularly pernicious and is compared to the act of Titus when he thrust his sword into the *paroches*. Asserting that man has no *bechirah* (free-will), that man lacks the ability to control his actions, amounts to denying that there is a *neshamah* present inside of him! This is akin to denying the presence of the *Shechinah* in the *Beis HaMikdash* and imagining that a mortal can inflict harm on Hashem.” (*Zichronos*, Harav Yitzchok Hutner, Rebbitzin Beruriah David).

In a letter (*Iggeres* 42), the Rosh Hayeshivah elaborated on the concept of denying *bechirah* and explained that “The absolutely worst manifestation of *tumah* (corruption and degradation) in our times is the **זלזול בצורת אדם**, the belittlement of man's stature.”

צורת אדם is not man's physical appearance, but the distinctive human form of man's elevated nature in the image of God as a rational, choosing being. Human greatness lies precisely in the capacity of intellect and free choice to recognize reality and act accordingly. Pharaoh's tragedy is not that he lacked intelligence; it is that he used his freedom to resist what his own intelligence—and even his own experts—should have compelled him to acknowledge.